# Slide-a-Key: Teacher's Guide

Slide-a-Key is a free, printable paper transposer. Students cut it out and build it, then use it to move notes and chords from one key to another.

The reason to use it in class is that the tool makes the structure of transposition visible:

- The strips show that transposing is a fixed distance. Move the slider once, and every note shifts by the same number of boxes.
- The wheel shows that the twelve notes form a loop. Go far enough past B and you're back at C.

Students can find both patterns with their hands before anyone names them.

## Why It Works in Class

- They make it. Cutting, folding, and fastening turn a worksheet into an object. Students own a tool they built themselves in a way they never own a handout, and the first time the notes line up, it's their own tool that worked.
- They discover it. Hold back the explanation. When a student notices that every note moved two boxes, they've found the idea of an interval for themselves, and people tend to remember what they work out for themselves longer than what they're told.
- They keep it. Let students write their name on the back, colour in the sharps, and take it home or keep it in an instrument case. A tool they reach for next week keeps teaching.

## Choosing an Edition

| Edition | Build time | Extra materials | Good for |
|---|---|---|---|
| Classic | About 5 minutes | None, though a ruler helps | A quick start; the slider stays lined up |
| Pocket | About 10 minutes | A brass paper fastener (split pin) for each student | A craft session; something to keep in a case or folder |
| Sleeve | About 20 minutes, plus drying time | A glue stick | Something sturdy to keep; the slider can't fall out |

All three teach the same core idea. Pick by the time and materials you have. Optional add-ons for each edition (free at slideakey.com) mark I, IV, V, and vi in any key. The add-ons highlight the main chords, which the transposer also finds, and they can find them on their own, without the slider.

## Materials

- One printed sheet per student, at 100% scale (not "fit to page"): the print-check line at the bottom should measure 50 mm
- Scissors: the only cutting tool needed for any piece, windows included
- Pocket Edition only: one brass paper fastener per student. Its point makes the centre holes.
- Sleeve Edition only: a glue stick
- Optional: cardstock or heavier paper, for sturdier copies

## A Sample Lesson (about 45 minutes)

1. Build it (5–20 minutes). Students follow the assembly steps printed on the sheet.
2. Try the example (5 minutes). Everyone moves C up to D, as in the printed example, and checks that the main chords (I, IV, V, and vi), C, F, G, and Am, become D, G, A, and Bm.
3. Explore (10–15 minutes). Work through some of the discovery questions below in pairs or as a class. Have students predict each answer before they move the slider, then check.
4. Use it (10 minutes). Transpose the chords of a song the class knows into a new key, then play or sing it.
5. Keep it (2 minutes). Students sign the back of their Slide-a-Key and take it with them.

## Discovery Questions

Ask these before explaining anything. The answers are for you.

1. Count the boxes between C and D. Now count between F and G, and between G and A. What do you notice? They're all two boxes apart. Transposing moves every note by the same distance, called an interval. Each box is a semitone (half step), so two boxes make a whole step.

2. What happens if you slide the slider 12 boxes, or turn the wheel all the way round? You're back where you started, an octave higher. There are only twelve different note names, and they repeat.

3. Why do the strips have every note twice, when the wheel has each note only once? Because the notes form a loop. A strip needs the repeat so you don't run off the end; a circle loops back on its own.

4. Which boxes are shaded? Where have you seen that pattern before? The sharps and flats, which are the black keys on a piano. Just as on a piano, there are none between B and C, or between E and F.

5. Am becomes Bm. G7 becomes A7. What changes in a chord name, and what stays the same? Every note letter moves, even a bass note after a slash (G/B becomes A/C#). The "m", "7", and "sus" stay.

6. Move C up 5 boxes. Now, starting again, move C down 7 boxes. What do you get each time? F is the IV chord in C: why 4, when it's 5 boxes away? F both times, an octave apart, because 5 + 7 = 12. IV counts the letters of the C scale, C D E F, so F is the 4th. The boxes count every note, sharps included.

7. Without an add-on, how could you find the I, IV, V, and vi chords of D? Of any key? Move C up to D: C, F, G, and Am become D, G, A, and Bm. Every chord name changes, but the numerals don't: they're still I, IV, V, and vi. Moving C to any key turns them into that key's I, IV, V, and vi. The add-ons find the same chords and highlight them, so they're easier to see.

8. For a song in A minor, put an add-on's blue mark on A (the left one, if it has two). Why is A labelled vi, when the song is in A minor? The numerals count from the green mark, which now sits on C. A minor and C major share all their notes, so a song in Am uses the same family of chords: Am, C, F, and G. Small letters mean minor, so vi is Am. Counted from A instead, the usual way to number a minor key, the same chords are i (Am), III (C), VI (F), and VII (G): blue is i, green III, yellow VI, and red VII. The website's Try Me uses these numbers when you choose minor; each add-on mark carries one label, so the marks keep the major-key numbers.

9. C is I in the key of C. What is it in the key of G? In F? IV in G, and V in F. A numeral names a chord's job in a key, not the chord itself. On the website, set the song to G and C appears as IV; with an add-on, put the green mark on G and C sits under the yellow.

## Things to Watch For

- Windows, with scissors. Cut the window before the piece. Fold the sheet on the dotted line across the window (the ticks above and below keep the fold straight), cut both long sides from the fold to the corners, then fold the flap down and snip it off. The steps are printed in each window too.
- If the boxes don't line up, check the build, not the printer: a crooked cut (Classic), an off-centre hole (Pocket), or a sagging slider (Sleeve). "Fit to page" only makes everything a little smaller.
- Spelling. Slide-a-Key gives the right sound, not always the textbook name: in F# major, written music calls F "E#". For chord charts and playing, the strips' names are fine. Moving to F, B♭, E♭, A♭, D♭, or G♭? Read the flat names.
- Minor keys. With an add-on, put the blue mark (the left one, if it has two) on the key's own note (A for a song in Am). The marks then show Am, C, F, and G, the chords many minor-key songs use, pop songs especially. Roman numerals count up from the key's own note: capitals mean a major chord, and small letters a minor one. In C, I, IV, V, and vi are C, F, G, and Am; in A minor, the same chords are i, III, VI, and VII. Older folk and traditional songs often use the key's own Dm and E (or Em) instead: put the green mark on A, read the roots, and ignore the blue. On the website's Try Me, choose minor beside the key, and it numbers the chords i, III, VI, and VII.
- Which part is which. The fixed strip or outer ring is the key you're in now. The slider or inner disk is the key you're moving to.

# Going Further

- Transposing instruments. Some instruments read music written in a different key from what sounds. To find the written note from the sounding (concert) note, move the slider up by:

| Instrument | Move up by |
|---|---|
| B♭ trumpet, clarinet, or tenor sax | 2 boxes |
| E♭ alto or baritone sax | 9 boxes |
| F horn | 7 boxes |
| Violin, piano, guitar, flute, or voice | No change |

  Slide-a-Key gives note names only, not octaves. Guitar, tenor sax, and baritone sax also read an octave higher than they sound, so the written note may sit in a different octave.
- Guitar capo. With a capo on fret N, move down N boxes to find the chord shapes to play. Going the other way, move up N boxes to find what a shape will sound like.
- Guitar-friendly keys. Move C up to G (7 boxes). C, Am, F, and G become G, Em, C, and D, and every one is an easy open chord. It's one reason so many guitar songs are in G.
- Violin-friendly keys. The open strings are G, D, A, and E, so G, D, and A major are often the easiest keys for beginners. Move a song into one of them; a melody's notes move just like chords.
- Piano-friendly keys. The white keys are the unshaded boxes on the strips, and C major uses only those. G and F add one black key each. Move a song into C, G, or F for a beginner.
- Singing range. Ask the class to find a key that suits their voices, then transpose the song into it.
- Try an add-on on its own (optional, if time allows in class, or as something to try at home). Leave out the slider or disk and use just the base and the add-on: the Chord Family Classic over the fixed strip (Classic), the Chord Family Pocket on the base card without the disk (Pocket), or the Chord Family Sleeve over the Sleeve with the slider removed (Sleeve). Move it to a key and read off I, IV, V, and vi. It still works: finding the main chords is one job of the transposer that an add-on can do by itself, and it highlights them too.
- Make the boxes bigger. Have students draw their own strips or wheel from scratch. If they get the pattern of shaded boxes right, they understand the chromatic scale.

# Sharing

Slide-a-Key is free for classroom use. You may print and photocopy as many copies as you need, and share the files with other teachers. Please don't sell it. See **`LICENSE.txt`** for details (CC BY-NC 4.0). More at www.slideakey.com.

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